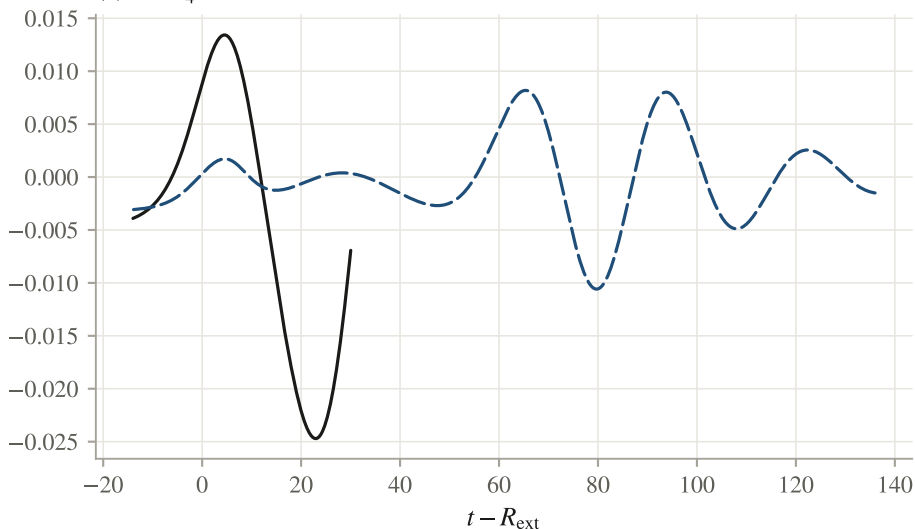
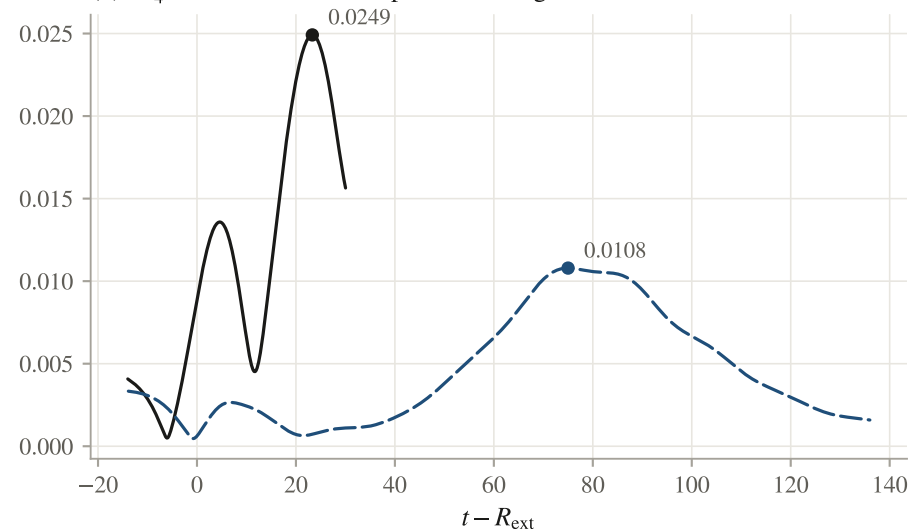


Same masses, same orbit ($d=12$, $p=\pm 0.12$, $M_{\text{ADM}}=1$ each) — different object

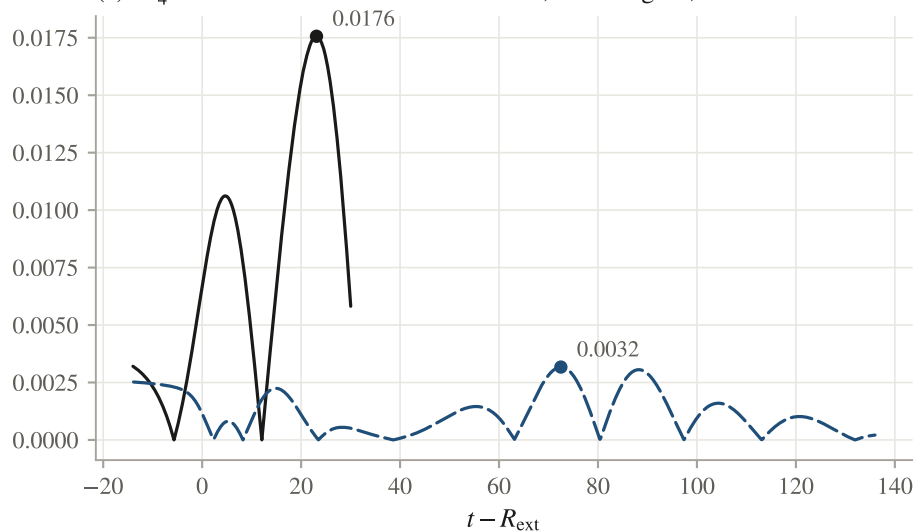
(a) $\text{Re}\Psi_4^{2,2}$ at $R=14$



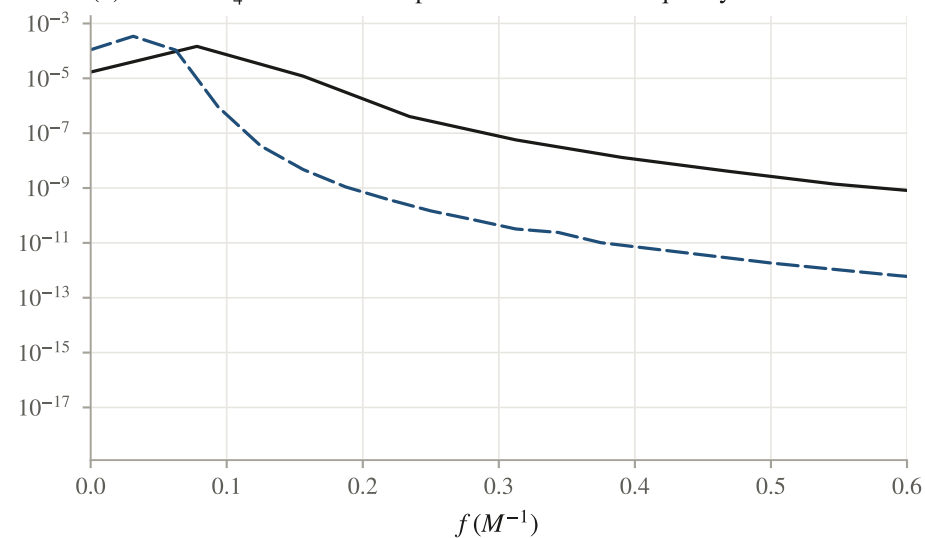
(b) $|\Psi_4^{2,2}|$ — the wormholes peak $2.3 \times$ higher, 52 earlier



(c) $|\Psi_4^{2,0}|$ — the wormholes' extra channel, $5.5 \times$ higher, 49 earlier



(d) PSD of $\Psi_4^{2,2}$ — wormhole power sits lower in frequency



— wormholes ($p=0.12$ twin)

— black holes (vacuum control)